

Correspondence

When productivity comes at the cost of apprenticeship

In the past few months, we have worked extensively with the artificial-intelligence assistants Claude Code and OpenClaw on various research projects. They have been the perfect assistants that any researcher would want: they are always available, can review literature, grasp our guidance quickly and can debug complex codes in minutes or hours, rather than weeks or months. It's tempting to outsource most data collection, cleaning and curation to these tools. We doubt we are alone in feeling this way.

But we have noticed something that troubles us. We have not assigned substantive tasks to our graduate research assistants in weeks. We still advise them on their papers, but when it comes to our own research agenda, we have quietly turned to AI agents instead. As a result, our meetings with members of our research groups have become less frequent, and students are less involved in our projects' research procedures. They know less than they might otherwise have done about the choices, dead ends and small calibrations that shape a project and lead to a product.

That realization gave us a jolt. AI saves time, but it also deprives students of the cognitive scaffolding they need to become autonomous scholars. As mentors to emerging scholars, we must balance the use of AI with the need to preserve the hands-on research experiences required for students' growth.

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Legal rights for insects

Stingless bees (*Meliponini*) are now recognized as legal entities in the municipalities of Satipo and Nauta-Loreto, Peru. They are the first insects in the world to hold legal rights, marking a turning point in environmental jurisprudence. Legislation grants native stingless bees the right to exist, thrive and be represented in court.

Stingless bees pollinate around 80% of tropical flora. For example, Ethiopia's diverse Afromontane forests rely on native species such as *Meliponula* to sustain wild biodiversity and pollinate crops on which the country's economy relies, including wild *Coffea arabica*, which accounts for up to 45% of its total export earnings.

But the bees' biomes remain dangerously unprotected. Insects remain highly vulnerable to habitat fragmentation and the indiscriminate use of agrochemicals. By empowering Indigenous groups and conservationists to sue on behalf of insects harmed by deforestation or pollution, legal systems can directly address the drivers of ecological collapse.

We urge policymakers worldwide, particularly those in biodiversity hotspots across Asia and Africa, to adopt biocentric legal models for safeguarding global food webs and preserving tropical biodiversity.

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Precision medicine without equity is just stratified inequality

Precision medicine holds immense potential to transform health care by tailoring prevention, diagnosis and treatment to each individual (see F. D. Urnov and S. H. Kassim *Nature* **652**, 857–859; 2026). Advances in genomics, large-scale data integration and artificial intelligence have accelerated progress. Yet without deliberate attention to equity, the approach risks entrenching existing disparities rather than reducing them.

Precision medicine relies on data. But genomic and clinical data sets remain disproportionately derived from populations with European ancestry. Risk prediction, for example, is often poor outside populations in which the models were developed.

Limited access to technology further compounds this problem. Advanced diagnostics and targeted therapies are concentrated in well-resourced settings. In high-income countries, access varies by geography, socioeconomic status and insurance coverage. Globally, the gap is even more pronounced. The growing integration of AI introduces further challenges: models trained on biased data might perpetuate inequities while appearing objective.

Addressing these risks is not just a technical issue. Equity must be embedded as a foundational principle through inclusive data collection, rigorous evaluation of model performance across populations and policies that ensure equitable access.

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Thymic health under the microscope

In a recent paper in *Nature*, Bernatz and colleagues report a measure of the health of the thymus – an organ in the chest that produces T cells. The researchers derived a 'thymic health' score from an analysis of computed tomography (CT) images, and showed that a low score was associated with mortality and several disorders (S. Bernatz *et al.* *Nature* **652**, 986–994 (2026); see also G. Anderson *Nature* **652**, 865–866; 2026).

The thymus is known to shrink and its function to decline with age. This study is purported to provide evidence that the organ retains clinical relevance throughout life, but we urge caution. It introduces an imaging biomarker that is itself strongly associated with age-related effects. Reduced thymic size is inferred to reflect smaller output, yet thymic shrinkage can be driven by illness and stress.

It is not clear whether the studied cohorts are free of co-morbidities. Monitoring systemic inflammation and taking into account age-related effects would have helped to identify thymus-specific biology. Conditions such as the autoimmune disease myasthenia gravis, in which thymic pathology is central (H. J. Kaminski *et al.* *Neurology* **102**, e209482; 2024), were not discussed. More longitudinal and mechanistic studies are needed.

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